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Edition No.: 106
RETROFIT SPECIAL
 Housing Recovery and
Reconstruction Platform
BULLETIN
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Singha Durbar, Nepal Government's main administrative building that housed federal ministries and departments including National Reconstruction Authority (NRA) suffered damage due to the 2015 earthquake.

Top: The Western Part of Singha Durbar. NRA has recently decided to retrofit and preserve the original structure of the building. **Bottom:** The eastern part of Singha Durbar undergoing retrofitting.

HRRP3 External Evaluation

As HRRP moves to its fourth phase starting from 1 August 2019, the final evaluation of HRRP3 is imperative. The final evaluation from the independent external evaluator is currently being conducted. The major objective of the evaluation is to measure relevancy, effectiveness, efficiency, impact and sustainability of HRRP3 interventions and to collect evidence based learning for HRRP fourth phase and other recovery and reconstruction stakeholders at

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national and international level. The respondents for this evaluation are central, district and municipal level government, reconstruction partners, donors and other international experts. The evaluation is expected to be completed by the third week of September 2019.

Retrofitting: a Solution for Resilient Recovery

The damage assessment survey has identified around 70 Thousand House that could be retained their seismic strength through retrofitting. With the start of the reconstruction phase most of the training and technical assistance systems promoted for the reconstruction were aimed at new construction only. with the approval of retrofitting manual by NRA for [masonry structure](#) as well as [RCC structure](#) on June 2017, it has been a turning point, with a clearly defined retrofit solution is made available to the large numbers of homeowners for the first time. Retrofitting solution could not attract the homeowners Over the last two years the way it has to, due to the lack of proper technical guidance to the engineers and masons in the field and communication gap from the NRA, CLPIUS and concerned GoN authorities to the affected areas on the benefit of retrofitting resulting to no interest from the beneficiaries and beneficiaries switching or planning to switch from retrofit to reconstruction causing delay in recovery/reconstruction. Retrofitting allows homeowners to return rapidly to their seismically strengthened home and to renew their agrarian lifestyle sooner whereas the change in the beneficiary type from retrofit to reconstruction results to more financial burden and debt for the demolition and reconstruction of new house to the families are having hard time in sustaining their livelihoods. In rural villages where houses are both a home and a farm usually consisting of a barn, a granary and a home, all in one. Retrofitting such a rural house not only protects families and their livestock from the impact of future earthquake damage, but also enables families to return to their pre-earthquake level of economic productivity, thus restoring livelihoods.

With the start of the retrofit engineers training and model house retrofit from some of the POs there is some progress in the retrofitting. The NRA and GoN is also keen to make the beneficiaries understand and take retrofit a feasible and cost-effective option for resilient recovery. Different sharing session has been organized to identify the issues and challenges in the field from the PO's as well as beneficiary's side. The workshop organized on July suggested the GoN a need of strategic intervention in policy revision, robust communication mechanism to the field and capacity enhancing of the field technical staffs. For detail contact Minar Thapa Magar, HRRP Deputy National Coordinator (minar@hrrpnepal.org)

Featured Technical Staff Person



Shiv Prasad Bhandari, District Support Engineer, Bhaktapur

This month's featured technical staff person is Shiv Prasad Bhandari. He has been working as a District Support Engineer (DSE) in Bhaktapur. Before being deployed as DSE he has worked with DLPIU Education and was directly involved in the retrofitting of a school building in Nawalparasi. He feels that motivating the community towards retrofitting is very challenging because there has been low communication and awareness about the benefit of retrofitting. The capacity enhancement of field technical staff is also key to implement the retrofitting in the community.

Shiv feels that it has been a very welcoming effort from the GoN as well as POs to train the field technical staff through retrofitting the houses as model house. This field demonstration and restoration of the previous structure will influence the community to willingly select the retrofitting as the viable reconstruction option.

Highlights

- HRRP Published [Reconstruction Facilitators at Local Level \(English and Nepali\)](#) a booklet to share the varied experiences of mayors, deputy mayors, ward chiefs, municipal chairpersons, and administrative and operation staff within District Level Programme Implementation Units (DLPIUs) in earthquake affected districts.
- [Retrofitting Sharing Session and workshop 29 July, Hotel Summit](#)
- [Asia Pacific Regional Shelter Cluster Workshop and Shelter Forum, Indonesia 21-23 August](#) Nepal organizing next Asia Pacific Shelter Forum in 2020
- 646 households, across sixteen districts, have switched from the retrofitting grant to the reconstruction grant
- Almost 60% of eligible households have received the third tranche of the housing reconstruction grant

National Updates

[NRA, CLPIU GMaLI and Building, Grant Disbursement Data, as of 26 August:](#)

Reconstruction Grant	17 June	15 July	26 August
Total No. Eligible HHs	826,333	826,634	816,253
HHs Enrolled	766,316	768,192	776,101
HHs Received 1 st Tranche	761,542	763,470	769,105
HHs Applied for 2 nd Tranche	598,295	607,597	613,964
HHs Received 2 nd Tranche	584,516	590,542	605,254
Non-compliant at 2 nd Tranche	4,287	4,277	4,234
HHs Applied for 3 rd Tranche	458,715	479,811	495,452
HHs Received 3 rd Tranche	436,272	453,354	487,957
Non-compliant at 3 rd Tranche	2,222	2,005	1,865
Houses Completed	417,395	418,760	428,513
HHs with Completion Certificate	16,027	24,361	24,739
Retrofit Grant	17 June	15 July	26 August
Total No. Eligible HHs	66,664	71,358	69,973
HHs Enrolled	21,587	21,984	33,468
HHs Received 1 st Tranche	20,877	21,583	31,512
HHs Received 2 nd Tranche	41	45	27



Progress with the disbursement of the GoN housing reconstruction and retrofit grants based on NRA, GMAI CLPIU data as of 26 August 2019.

Retrofitting Sharing Session and workshop 29 July, Hotel Summit: during the session organized by NRA, Building CLPIU, UNOPS, UNDP, NSET, and some of the beneficiaries shared details of challenges and opportunities in relation to retrofitting. The workshop focused on discussing on the major issues currently prevalent in the successful implementation of retrofitting. The participants were divided in 4 different groups each group was led by NRA Executive committee members each to discuss the major challenges and provide the strategic solution to speed up the retrofitting. Closing the session, NRA CEO urged the need of enhancing the technical capacities in the municipalities with proper training package to engineers and masons with suitable communication and social mobilization where the homeowners and local communities are more informed and are able to reconstruct their houses based on the need.

Partners Sharing Session with GoN on the Housing Reconstruction Progress, 14 August, DUDBC: the objective of the event, which was organized by NRA with the coordination support from HRRP was to review housing progress of the partner organizations working in housing sector and interact with NRA Officials on housing reconstruction issues & challenges, lessons learned and way forward for the future. 13 implementing POs presented their progress during the event. Closing the event NRA CEO urged the POs to work closely with the local government capacitating them in communication, outreach of the information along with proper documentation and development of knowledge products for future.

Technical Support to Lumjung and Tanahu GMAI and Building DLPIUs, 12-16 August, HRRP: In accordance with by CLPIU-GMAI, HRRP national IM team visited Tanahun and Lamjung with the objective of providing the technical support to the DLPIUs for data uniformity and cleaning in line with the existing MIS. The data management systems were not well established in the moderately affected districts where computer operators with minimal guidance and orientation are responsible for the data management. During the event HRRP also piloted a system developed to access the tranche and inspection status of beneficiaries. Orientation regarding the information system was also provided to the computer operators of Building and GMAI in both Lamjung and Tanahun districts. The data records in DLPIU Building and GMAI in both the districts were migrated to the new system and they are already using it. The concerned staffs (largely the computer operators) were very happy to get the improved Information Management system because the system with the inbuilt search functionality allowed them to answer the queries of beneficiaries on their tranche status almost instantly. For detail contact Yubaraj Bhandari National Information Management Officer (nimo3@hrrpnepal.org)

Asia Pacific Regional Shelter Cluster Workshop and Shelter Forum, Indonesia 21-23 August: The Regional Shelter Cluster Workshop provides an opportunity for regional shelter cluster stakeholders in Asia-Pacific to contribute to Global Shelter Cluster Working Group initiatives and to share practice on shelter cluster coordination. Whereas the Asia-Pacific Shelter Forum brings together practitioners to share ideas and lessons learned on humanitarian shelter and settlements initiatives and programs. The HRRP Deputy National Coordinator took part in the event and shared

the experience of Nepal's housing recovery after the Gorkha earthquake 2015 in a session on country wise experience sharing. The presentation is available [here](#). The Forum decided to hold next Asia Pacific Shelter Forum 2020 in Nepal. **Experience Sharing Event on Skills for Safe Reconstruction, 22 August:** the event was organized by Swisscontact to share the experience of the Skills for Safe Reconstruction Project (SSRP) that was implemented in Sindhuli from 2016 till 2019. The sharing event highlighted some of the best practices of the project such as capacity development of the HHs to understand their awareness on safe reconstruction along with the skill enhancement of construction workers. The event was attended by NRA executive member, Building CLPIU, Local Government and other partner organizations. Addressing the event Dr. Chandra Shrestha from NRA, highlighted the importance of coordination of the implementing Partners among themselves and with the local governments to reduce the gap and duplication in reconstruction support. He appreciated the workshop and urged all the POs to organize similar events and publish the best practices and learning documents for future.

Engineer's Training on Seismic Retrofit Design of Masonry Buildings 19-23 August: the training was organized by NSET Baliyo Ghar with the objective of enhancing the capacities of technical personnel, especially structural engineers and designers to undertake the design of retrofitting of masonry buildings abundant in rural Nepal. The training covered a wide range of theoretical and practical sessions involving vulnerability and damage assessments of existing buildings, case studies of retrofitting in masonry buildings and manual and SAP based modelling, analysis and design of retrofitting. Total 30 engineers working in the reconstruction have participated in the training. NSET Baliyo Ghar has organized 3 batches of similar training in past which has trained 89 Engineers in seismic retrofit design.

Earthquake 2015 - Open Data Portal' inauguration programme 26 August: Following the 7.8 Mw Gorkha Earthquake on April 25, 2015, the Central Bureau of Statistics (CBS), under the guidance of the National Planning Commission (NPC) and with technical support from Kathmandu Living Labs (KLL), had conducted a massive door-to-door housing survey to help the National Reconstruction Authority identify beneficiary households. CBS had previously released the anonymized data as [Earthquake 2015 Microdata](#) for public use. Realizing that this dataset is not only useful for reconstruction but also for other planning and development purposes both at local and national levels, NPC supported KLL in the development and launching of '[Earthquake 2015 - Open Data Portal](#)'

Nepal People's Forum on Sustainable Development 2019: Nepal People's Forum on Sustainable Development 2019 was held on 22 August at Alfa House in Kathmandu with the theme "Accelerating civil society engagement: Realizing Social Development Goals (SDGs) in action." The overall objective of the forum was to discuss on key development issues particularly related with upcoming UN General Assembly taking place in 24-25 September 2019. The event was participated by over 100 participants comprised of NGOs and CBOs, youth representatives, persons with disability (PWD), LGBTIQ, government officials, including representatives from National Planning Commission and Ministry of Foreign Affairs among others.

District & Municipal Updates

Number of households switching grant type: this data has been collected from the DLPIU offices by the HRRP District Coordination Teams. It will be updated as more information is received.

District	No. of HHs changed from Retrofitting to Reconstruction	No. of HHs changed from Reconstruction to Retrofitting
Dhading	7	0
Dhankuta	25	0
Dolakha	21	0
Gorkha	71	0
Kathmandu	1	5
Kavrepalanchok	154	5
Khotang	4	0

Lalitpur	16	0
Makwanpur	219	17
Nuwakot	25	0
Okhaldhunga	22	0
Ramechhap	2	0
Rasuwa	39	0
Sankhuwasabha	8	3
Sindhupalchok	24	0
syangja	2	3
Total	646	33

HRRP District and Municipal Coordination Meetings, 15 July – 23 Aug 2019: during this period district coordination meetings took place in Kathmandu (7 Aug). Municipal level meetings took place in municipalities and wards across Dolakha, Gulmi, Kathmandu, Kavrepalanchok, Kaski, Lalitpur, Lamjung, Palpa, Ramechhap, Sindhuli & Sindhupalchok. Some of the highlights from these meetings include:

In **Kathmandu**, district facilitation and coordination meeting was held on 7 Aug 2019. Some of the decisions made in the meeting were: to find the households that have taken grant from the Government and any partner organization, and the households that are not able to start reconstruction due to land issues. DLPIU Education shared that it will list out the municipalities that need to be coordinated for the provision of land for school reconstruction and submit it to District Coordination Committee (DCC). GMaLI to coordinate with all the banks and concerned stakeholders to conduct meeting to resolve the issues related to banking sector that have reported in Kathmandu. Further, GMaLI to coordinate with municipal and ward representatives either to bring the households that have not signed the partnership agreement into the Government reconstruction process or to remove them from the list.

In **Gulmi, Gulmidurbar Rural Municipality (17 July 2019) and Ruru Rural Municipality (17 July 2019)**, the meetings discussed on the non-compliant houses constructed immediately after the 2015 earthquake, households that received the first tranche and have not started construction and issues related to conversion of households from retrofit to full reconstruction grant list. The meeting further discussed on retrofitting trainings and demonstration to show and encourage the households to carry out retrofitting. In Ruru RM, 66 households have received reconstruction completion certificate.

In **Palpa, Tansen Municipality (18 July 2019) and Bagnaskali Municipality (19 July 2019)**, the Mayor of Tansen Municipality, raised concerned regarding the reconstruction of temples and old VDC buildings which have not been reconstructed yet and no stakeholders showing interest as well. Tansen Municipality office has provision that allows earthquake affected households to pay only 50% of the tax on building permit. Technical staffs in the field also requested for retrofitting trainings.

In **Kathmandu, Tokha Municipality (1 Aug 2019)**, the meeting focused on non-compliant houses that were constructed immediately after 2015 earthquake before the deployment of technical persons by NRA. Most of the houses have used 9"x12" column size, missing earthquake resistant elements in the construction. There are few people who have not been able to sign the partnership agreement to be get the Government financial assistance to start reconstruction due to issues with landownership documents and road extension program.

In **Kathmandu**, general orientation and discussion program focusing on the private housing reconstruction were conducted in **all the municipalities from 5 August – 21 August 2019**. Mr. Hari Prasad Sharma, Project Manager, DLPIU-GMaLI, Kathmandu presented the status of the reconstruction of Kathmandu district. He mainly focused on the challenges and issues that has been affecting the reconstruction process. He further mentioned that NRA has made the policy for allowing the reconstructed in all the previously restricted land such as government, forest area, public land, trust land and so on. Mr. Bidhya Sundar Shakya, Mayor, Kathmandu Metropolitan City in his remarks highlighted that the reconstruction especially the core areas of not only Kathmandu Metropolitan City but

Kathmandu Valley must have different modality as compared to reconstruction of the other districts. District Coordination Committee Chief highlighted that with the effort of all the concerned sectors it's time to visit each beneficiary and encourage them to reconstruct the houses or remove the fake beneficiaries from the list.

In **Lalitpur**, meetings took place in **Ward 3, 11 & 16, Lalitpur Metropolitan City (26 July 2019), Ward 21 (9 Aug 2019) & Ward 2 (12 Aug 2019)**. People having houses in other places, dispute over land ownership among family members, small parcel of land, families residing in multiple storey in the same house, removal of rubbles in the core areas due to narrow road are reported as common reconstruction challenges in LMC. In ward 21, the main challenges are maintaining the cultural authenticity of heritage settlement and lack of proper awareness about building bylaws of heritage settlement. Ward office suggested that technical personnel with knowledge and skills on construction of heritage settlements would be helpful for reconstruction in heritage sites. It was noted that additional 100,000 NPRs has been provided by LMC for houses reconstructed in heritage settlements. Regarding retrofitting, people do not have proper technical assistance even they intend to retrofit their houses.

In **Ramechhap** meetings took place in **Ward 4, Manthali Municipality (6 Aug 2019), Ward 3 Manthali Municipality (7 Aug 2019), Ramechhap Municipality (8 Aug 2019) & Ward 7 Ramechhap Municipality (9 Aug 2019)**. The main reason people are not showing interest in retrofitting is the lack of knowledge and understanding. A model house, demonstrating retrofitting techniques and processes would be a good way to motivate the people in retrofitting was suggested during the meeting. Most of the houses under retrofit grant repaired their houses, cracks are not visible now so, this has caused challenge for the technical staffs to carry out the work. Many of the households in retrofit grant have not signed the PA and have applied to be converted to be in full reconstruction grant list.

In **Sindhuli, Golanjor Rural Municipality (6 Aug 2019) & Marin Rural Municipality (8 Aug 2019)**, it was discussed that 570 PA forms from Golanjar RM and 920 from Marin RM are remaining to be collected and send to DLPIU Building for digitization. The meeting shared the decision on 2,615 landless households so far that could start reconstruction in ailani land (government land) where they have been living for generation but did not have any land ownership or legal documents.

In **Kavrepalanchok** meetings took place in **Panchkhal Municipality (7 Aug 2019), Roshi Rural Municipality (9 Aug 2019), Namobuddha Municipality (13 Aug 2019), Bethanchok Rural Municipality (19 Aug 2019)**. The common challenge reported was gap in communication at ward level, most information relating to reconstruction policies/guidelines, deadlines and other notices are either not reached or reached late in wards. It was suggested that there should be proper communication mechanism for timely information flow in wards as the wards are first point of contact for the people and they are responsible for implementing major activities and to act. Many of the households in retrofit grant that applied to be switched to full reconstruction grant are still waiting for the decision, this has caused delay in reconstruction. Confusions to the applicants remains due to delay in grievance handling despite timely appeal done. A long time taken in changing the beneficiary name in the event of death of initial beneficiary was also discussed during the meeting. In Panchkhal Municipality, building permits have been made mandatory to access the third tranche, this might delay the application for the third tranche. It was reported that the third tranche progress in Namobuddha Municipality is only 35%. In the meetings, it was agreed that DLPIU GMAI would coordinate with NRA to find out the status of households that have applied to be switched from retrofit to full reconstruction grant list. Similarly, DLPIU GMAI to coordinate with NRA to conduct appeal related specific programs at ward level.

In **Sindhupalchok**, meetings took place in **Ward 2, Helambu Rural Municipality (3 Aug 2019), Sunkoshi Rural Municipality (12 Aug 2019) & Bhotekoshi Rural Municipality (13 Aug 2019)**. In **Helambu RM**, main decision made was agreeing to conduct remeasurement of the land by concerned land measurement office as the land measurement done in March 2018 was horizontal instead of vertical in the process of purchasing land for the 6 displaced households of Helambu Rural Municipality ward no 2, Kiwool. Regarding the conversion of households from full reconstruction to retrofit grant, a technical report will be submitted to CLPIU for further amendment and final decision. In **Sunkoshi RM**, PA collection is going on, 1200 out of 21,000 PA have been collected which will be sent to

District GMAI for digitization. In **Bhotekoshi RM**, it was reported that there is misinformation in the community that the households under retrofit list would receive full retrofit grant of 100,000 NPRs at once while there is no such provision.

In **Dolakha**, meetings took place in **Baiteshor Rural Municipality (20 Aug 2019)**, **Ward 6, Jiri Municipality (21 Aug 2019)** & **Tamakoshi Rural Municipality (23 Aug 2019)**. The common issue reported was lack of understanding about retrofitting among local authorities and communities. People generally compare the grant amount between reconstruction and retrofit and are interested to construct new house instead of retrofitting their damaged house.

In **Lamjung**, **Marshyangdi Rural Municipality (19 Aug 2019)**, 161 households in retrofit grant have taken first tranche but have not started any work related to retrofitting. It was discussed that local masons are not confident to build resilient and safe structures due to lack of proper mason training. Regarding NRA decision to provide 300,000 NPRs housing reconstruction grant to the houses built before the deployment of technical persons in the field, has created confusion on minimum requirement and grant disbursement mechanism among the communities.

In **Kaski**, **Machhapuchhre Rural Municipality (23 Aug 2019)**, the meeting mainly focused on retrofitting cases and the ways to move forward with retrofitting. Procedures for the conversion from retrofit to reconstruction grant for houses with higher damage grade was discussed during the meeting. A task force was formed in Machhapuchhre Rural Municipality in coordination with DLPIU Building for implementation of retrofitting.

[HRRP District Coordination Team Bhaktpur and Lalitpur, Monthly Report, 15 July – 22 August 2019](#)



Ward 4, Mahankal Rural Municipality, Lalitpur: single storey, load bearing, stone in cement mortar structure with an attic. The home owner has received second tranche of the by GoN housing reconstruction grant and has applied for the third tranche.

HRRP District Coordination Team Kathmandu, Monthly Report, 15 July – 22 August 2019



Ward 6, Tokha Municipality, Kathmandu: RC frame structure. The home owner has received all three tranches of the Government housing reconstruction grant.

HRRP District Coordination Team Kavrepalanchok, Monthly Report, 15 July – 22 August 2019



Ward 4, Bethanchok Rural Municipality, Kavrepalanchok: single storey, stone in mud mortar house with an attic. The home owner has received all three tranches of the Government housing reconstruction grant.

[HRRP District Coordination Team Sindhupalchowk, Monthly Report, 15 July – 22 August 2019](#)



Ward 2, Helambu Rural Municipality, Sindhupalchok: single storey, two room, load bearing stone in mud mortar house with CGI roofing. The home owner has received all three tranches of the Government housing reconstruction grant.

[HRRP District Coordination Team Dolakha, Ramechhap and Sindhuli Monthly Report, 15 July – 22 August 2019](#)



Ward 4, Manthali Municipality, Ramechhap: single storey, two room. stone in mud mortar structure. The home owner has received all three tranches of the Government housing reconstruction grant. The total cost of construction is 600,000 NPRs.

[HRRP District Coordination Team Nuwakot and Rasuwa, Monthly Report, 15 July – 22 August 2019](#)



Ward No.2, Tadi Rural Municipality, Nuwakot: two storey, stone in mud mortar house with attic. The home owner has received first tranche of retrofit grant. The house was retrofitted with technical support from OXFAM.

[HRRP District Coordination Team Lamjung, Tanahun, Kaski and Syangja, Monthly Report, 15 July – 22 August 2019](#)



Ward No.1, Rainas Rural Municipality, Lamjung: two storey, three room with attic stone masonry in mud mortar house. The house is under retrofit grant list. The home owner has applied to be converted into the full reconstruction grant list. He is planning to return the first tranche of retrofit grant to the Government if the status of the house is not changed, and construct a new house after dismantling this old one.

Case Studies

Featured Technical Staff Person, Shiv Prasad Bhandari, this month's featured technical staff person is Shiv Prasad Bhandari District Support Engineer (DSE) of Bhaktapur. He is originally from Morang district. He has been serving as DSE for last one year. Before coming to Bhaktapur he was the DSE of Kavrepalanchowk district. Before being appointed as DSE he was working in the DLPIU Education which makes him well versed in the reconstruction and recovery of Nepal after 2015 Earthquake.



Shiv believes that retrofitting mostly deals with the technical corrections/modification in the existing damaged house hence most of the beneficiaries and other local house owners are not confident enough to choose retrofitting for reconstruction even after the suggestion from the technician. He also shared that the positioning of different structural elements within the house and Variation in Wall Thickness limits the retrofitting of the house.

He shared that he has been working continuously in the construction from 2002 which has contributed largely to provide the technical solution in the field. Apart from construction he was also directly involved in the retrofitting of a school building in Nawalparasi which helped him in polishing his technical knowledge and expertise in retrofitting. Besides the technical know-how, his participation in many trainings and orientation of retrofitting as participant and Facilitator has boost his facilitation and communication skills which

largely helped him in interacting and updating the affected families in the field.

Over the year he has found significant effort from the GoN as well as implementing Partners to make the retrofitting a viable and feasible reconstruction solution through different seminar, trainings, sharing session and media coverage which is very important for establishing a common understanding between all the reconstruction stakeholder. The training of the field technician including engineers and mason on retrofitting through the model house retrofit is a very good approach where the capacity of the technician is enhanced restoring the previous structure at the same time. However, he feels that the communication to the locals will be more effective if the local representatives could explain the benefit of retrofitting to them similarly the people believe what they see so it is important and imperative for the POs and GoN to involve the local more in demonstrating the retrofitting process during trainings and model house retrofitting along with the capacity enhancement of field technical staffs.

He is happy that he can share the technical knowledge and expertise he has in the earthquake affected communities to make the houses more seismically resistant and contribute to the national goal of build back better.

Bibek Bandhu Bhandari Quality Assurance Engineer, UNOPs

Bibek Bandhu Bhandari is a Quality Assurance Engineer for Retrofitting of SMM and Timber houses at Build Change. Graduated in Civil Engineering from Kathmandu Engineering College, he has been working in the field of engineering since three years and for the past 2.5 years, he has been working in retrofitting houses damaged by the earthquake.

He initially worked for shelter projects of World Vision, ARSOW Nepal and Build Change as Retrofitting Site Supervisor in Sindhupalchok district where he supervised retrofitting of 10 SMM houses. He then supervised five retrofitting houses in different rural municipalities of Kavrepalanchok district under Seismic Retrofitting of Unsafe Housing in Nepal Project of UNOPS/Build Change. He also worked for Quality Assurance of Model Houses in five different municipalities of Kavrepalanchok; Namobuddha NP, Dhulikhel NP, Temal GP, Roshi GP and Bethanchok GP. Apart from full partner supported retrofits, he coordinated in gathering interest and



thereafter retrofitting of 19 houses at MahankhalChaur, Bethanchok-6 with 70% support from UNOPS/Build Change. Currently he is working in Quality Assurance of Retrofitting of Timber Houses in Sindhuli district and SMM houses in Ramechhap and Sindhupalchok district.

Apart from the direct field experience, knowledge gathered through various trainings too supported his retrofit experience. He participated in Retrofitting Training on Major Interventions and Quality Control Training conducted by Build Change. These trainings have been useful in all his tasks of supervision of retrofitting houses, to make various interventions at field and quality control. Working in the retrofitting sector, he has had the chance to explore multi sectors of retrofitting like: Supervision, Mason Training, Training to NRA technical persons and Construction of Model Retrofit Houses.

Having worked in various retrofitting sectors for a long period, he also has experienced positive change in perspective and awareness level of communities towards retrofitting. Initially, beneficiaries, ward officials as well as NRA deployed technical persons didn't have positive impressions about retrofitting. Convincing them about the construction procedure and final strength of retrofitted house was a major challenge. But after several mass awareness programs to the communities and trainings to the NRA Engineers, the positivity was gained. After they have witnessed the retrofitting themselves, they are convinced enough.

As about 80% of houses in rural Nepal are under SMM typology, he believes that Pre-Earthquake and Post-Earthquake Retrofitting of all those houses is necessary and thus retrofitting has a wider scope. He thinks the lack of awareness in locals, insufficient number of skilled manpower and technical expertise for design, lack of coordination and concern of Government, insufficient support from Government to retrofitting beneficiaries pose challenge to the current retrofitting scenario and thus all these factors need to be considered to complete post-earthquake retrofitting.

Rudra Bahadur Regmi, Dhulikhel Municipality,

Rudra Bahadur Regmi is permanent resident of Dhulikhel-11, Patleket. His two story SMM house was damaged by the 2015 Gorkha Earthquake. He is one of the 5680 retrofit beneficiaries of Kavrepalanchowk district. Though he was listed as the retrofit beneficiaries but due to the lack of proper information about the benefit of retrofit and comparatively smaller grant support than that of the reconstruction grant from the GoN, he was intending to change from retrofit beneficiary to reconstruction beneficiary. But when he went to Manedobhan, Ryalebhir of Panauti he saw the retrofitted house of his relative and came to know about the benefit of retrofitting and its advantage in restoring the architectural design of the house in comparatively low cost. That event changed his perspective and he was convinced to retrofit his house. He participated in the UNOPS/Build Change Consortium launched "One Retrofit Demo House per Palika" program. His house is being retrofitted as one of the demo house where all the technical and material support was provided by UNOPS/Build Change but the beneficiary has to provide the financial contribute for the roof colouring. Now when he has received the PO support to retrofit house, he is planning to return the first tranche amount that was transferred to his account after the signing of PA. The total cost of the retrofit was 6.5 lakhs NPR



Tolak Bahadur Karki, Manthali Municipality,

Tolak bahadur is the permanent resident of Manthali Municipality Ward No. 2 kathajor, Ramechhap. His house sustained several cracks during 2015 earthquake. He was identified as the retrofit beneficiary by the damage assessment survey. He recalls that there were plenty of confusions early regarding retrofitting technique. Even he was not sure about retrofitting his house because he assumed that the retrofitted house is not safe like the one reconstructed with proper earthquake resilient elements. Through his neighbor and other local of his village, he came to know about Build Change which conducts repair and retrofit of minor damaged houses. With the motivation and support from the local and technical person from build change he was convinced to do the retrofitting of his house and contacted the Build Change Office. Initially he has agreed to procure the materials himself but his wife has kidney

failure which leads him to spend the saving for her treatment. Considering his economic status, Build Change covered all the cost associated with the retrofitting of his house.

Tolak Bahadur is well satisfied with the retrofitting of his house. He thinks that the house has become stronger than before. He is happy that he could restore his old house which has his emotions and past attached with it. He is thankful to Build Change for the support he has received. After his house was completed as retrofit model house, several beneficiaries of Kathajor visited to see and examine the retrofitting technique, even though they are excited and happy about restoring the previous building but are discouraged when they came to know that the costs associated with it is much higher than that supported by the government.



Left: Tolak Bahadur in his retrofitted house Right: The Retrofitted house Belonging to Tolak Bahadur Karki The retrofitting of the house was completed in 50 days and the total cost was NPR. 550,000

Gita Kumari Dahal, Deputy Mayor, Bidur Municipality: Gita Kumari was elected as the deputy mayor of Bidur



Municipality, Nuwakot on 2017. Before being elected as the local government representative she has supported the earthquake affected communities during relief and response phase too. After her election as deputy mayor she got chance to know and study about the issues and challenges more closely. Majority of the reconstruction beneficiaries have completed the reconstruction of their house in her municipality but there seems almost no progress in the retrofitting. Being the representative of the people, she believes that the identified beneficiaries should have to be included in the reconstruction beneficiaries because she believes that retrofitting is not a feasible and cost-effective option for the HHs in her municipality.

She both as an individual and the peoples' representative has not seen nor received any orientation/information that could motivate her to go to the community and encourage them to opt for retrofitting. The technical personnel (engineers/sub engineers) deployed in the fields are also not very confident and sure about the retrofit techniques and their importance in making the houses

seismically resilient. She has heard about some POs organizing the retrofit training for engineers and mason and doing the retrofitting of the houses as model retrofit house as part of the training program but such programs are not sufficient in the districts to motivate other beneficiaries. With the very little available information on retrofitting

as the appropriate and viable solution, the affected families are opting for reconstructing the houses that are smaller than the previous ones. Such small size houses are not able to serve the functional needs of the families of the village. She believes that if the GoN and associated Technical human resource are capacitate enough to communicate about the advantages of retrofitting and provide the required solution confidently to the affected families then the communities will be willing to opt retrofitting as the best solution for the seismic resistance of their partially damaged houses.

Incremental Retrofitting A Way Forward, Case study of UNDP:

Housing reconstruction is taking rapid momentum with robust policies and system in place, that have evolved in due course of time. Amidst reconstruction of new houses, we also cannot ignore the issues that await to be addressed systematically and pragmatically. For instance, many houses reconstructed preceded NRA guidelines and do not have earthquake safety features. Also, the old '*damaged*' houses are still being used besides the new house constructed with the reconstruction grant. While demolition of these old houses is mandatory, yet is a challenge economically, politically and comes with environmental implications.

In addition, the government has identified about 70,000 houses for retrofitting across 32 earthquake districts, while about 60% of the housing stock in Nepal are of similar nature, that is vernacular structure with mud-mortar based construction. It implies that more than 2 million highly vulnerable houses of similar nature are spread across the country, awaiting next disaster. In these vulnerable buildings, the risk of destruction poses the single greatest threat in a major earthquake leading to casualties, injuries and loss of property.

It is also evident that with such a huge scale, it is not possible to treat each structure separately to generate retrofitting/strengthening measures. Also, initial cost, lack of financial resources, lack of appropriate/affordable technologies and skills particularly on these vernacular structures, failure to realise risk and low financial return have been considered major obstacles for implementing seismic retrofitting.

In the given financial limitation, it is not feasible to retrofit majority of existing vernacular buildings to meet the Building Code requirements at the fullest, for achieving optimum level of safety, with minimal intervention and at the affordable cost. It is because, the building codes are mostly written for modern and strong materials and do not fully address the inherent problems of the vernacular buildings in entirety.

However, the technical fraternity needs to believe that these buildings still could be made safer, with the **philosophy of seismic design that assures life safety** against future earthquakes. The solutions have to be evolved that are replicable, affordable, viable and meets owners' aspirations.

To make retrofitting affordable, it should be acknowledged that it requires some level of trade-off between safety and affordability for reducing seismic risk to an acceptable level. Making it possible to select the elements that require retrofitting for addressing the basic seismic vulnerabilities first. This is an **incremental approach** that undertakes surgical approach to identify and address the most vulnerable elements and failure mechanisms first, followed by less vulnerable elements later as the funds become available over an extended period. It should also be noted that **addressing basic vulnerability delays the collapse thereby potentially protecting the human life**. Incremental retrofitting has been accepted and implemented by other countries exposed to similar level of seismic risk, and its credibility been tested with time and incidences.

Hence, a paradigm shift is necessary at this point, when we cannot afford to lose any time in the face of next earthquake. The target should be to reduce basic vulnerability first, that assures life safety and incrementally secondary vulnerabilities could be reduced to meet set level of safety in stages. Further to treat the vernacular buildings in massive number with similar failure modes and categories, the solutions for incremental retrofitting have to be generic leading to development of Mandatory Rule of Thumb for retrofitting.

Media coverage and communication on Retrofitting

This section highlights the past media coverage and materials produced and broadcasted with the objective to informing the community on the retrofitting and associated technologies.

[KathaMaala Radio Drama](#) KathaMaala is produced and broadcasted by Radio Nepal and BBC Media action with the financial support from DFID. The radio drama emphasis the importance of retrofit on restoring the original architectural structure within the affordable cost.

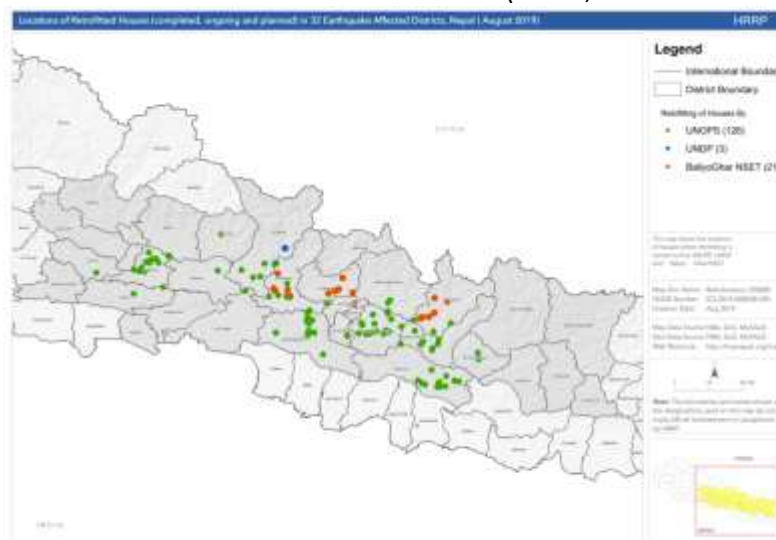
[Sajha Sawal Episode 548](#) this episode features Dr. Chandra Bahadur Shrestha NRA Executive Committee Member and highlights retrofitting as one of the GoN's priorities to complete the reconstruction within the stipulated deadline of NRA.



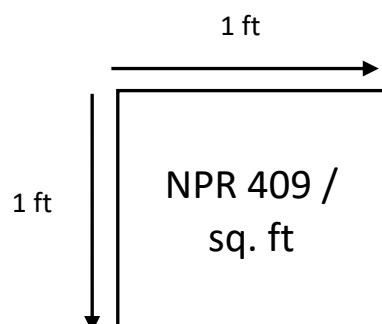
[Baliyo Ghar Episode 69](#)
[Baliyo Ghar Episode 132](#)
[Baliyo Gar Episode 138](#)
[Baliyo Gar Episode 139](#)
[Baliyo Ghar Episode 151](#)
[Baliyo Ghar Episode 152](#)
[Baliyo Ghar Episode 154](#)

HRRP Products

[Partners activity on retrofitting in the Earthquake affected districts:](#) HRRP has asked information on the retrofitting activities from Partners working in different earthquake affected districts. This map represents the ongoing retrofitting work based on the information received from Partners (UNDP, UNOPs and NSET)



The information received suggest that the unit cost of retrofitting varies across the housing typologies and the among multiple retrofitting solutions being implemented by the partner organizations. **The cost of retrofitting averaged NPR 409 per sq. ft.** (n = 150).



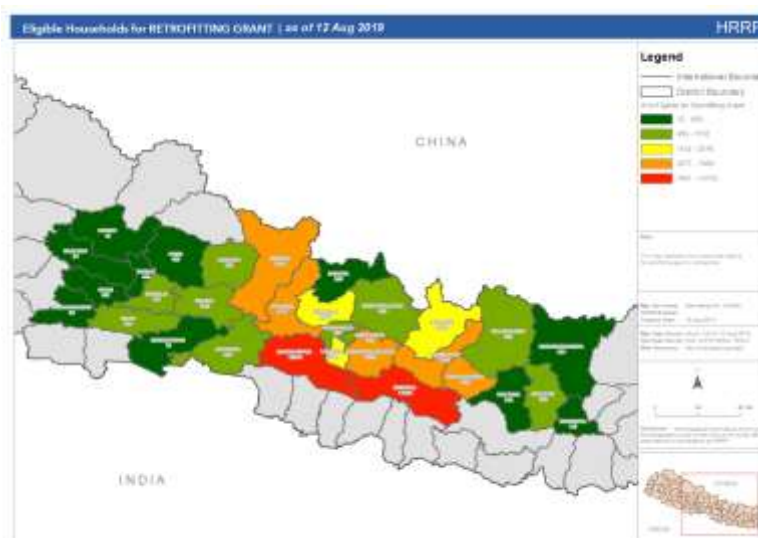
Housing Typologies	Avg. cost per sq. m (NPR)	Avg. cost per sq. ft (NPR)
Load Bearing Stone and Mud Mortar Masonry	6,208	577
Unreinforced Stone masonry in mud mortar	4,608	428
Unreinforced Dry Stone Wall	4,557	423
Brick in Cement Mortar	4,029	374
Hybrid Structures	2,631	244
Average cost of retrofitting	4,407	409

Partner organizations reported implementing retrofitting solutions for a total six housing typologies:

- Load Bearing Stone and Mud Mortar Masonry
- Unreinforced Stone masonry in mud mortar
- Unreinforced Dry-Stone Wall
- Brick in Cement Mortar
- Hybrid Structures
- Timber frame structure

Among the houses being retrofitted, more than 74% of the houses have two or more storeys and about 72% of the houses have attic. The most common methodology in retrofitting implementations were found to be strongback and split and bandage with numerous variations.

Retrofit Caseload Map: HRRP has produced a map based on the number of eligible household present in 32 earthquake affected districts for retrofitting grant. The caseload numbers are based on the data from GMaLI CLPIU



[Reconstruction Facilitators at Local Level \(English and Nepali\)](#) : this book looks to share the varied experiences of mayors, deputy mayors, ward chiefs, municipal chairpersons, and administrative and operation staff within District Level Programme Implementation Units (DLPIUs) working on the reconstruction with the aim of highlighting the importance of their work, the challenges they have faced, and the solutions that have been developed to address these. This document includes representatives from all fourteen of the districts most affected by the earthquake as well as from Khotang, Myagdi, and Lamjung in the eighteen moderately affected districts.



Partner Products

[NeKSAP Nepal 2019 Flood Assessment Report](#)

[KathaMaala Radio Drama Episode 3:](#) Rachana Rasaili wakes up with a bad dream where earthquake devastates their house. She wakes up and writes a letter about her dream as she could not contact her mother on mobile phone. On the other side her mother reads her letter and writes back about her mobile being damaged and to build a two-room new house as their old house is partially damaged. Rachana replies and sends her mother a phone and writes that she consulted with her engineer friend who suggested her to retrofitting as it will be cost effective to retrofit rather than build a two-room new house. But they could not communicate as the mobile network tower in the village was broken. But they send letters back and forth. Rachana also asked her mother to visit the ward office to take assistance of the engineers. At last, Rachana's phone rings and she find out that her mother retrofitted their house.

[Baliyo Ghar TV, Episode 158](#) this episode highlights the five major challenges of reconstruction such as: retrofitting, vulnerable families, Urban Reconstruction, additional room and story in the existing house and beneficiaries who have not started reconstruction after receiving the first tranche.



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